

ASTHA KESHARI

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SUMMARY

Computer Science undergraduate (Data Science & AI) at IIIT Ranchi with 1+ years of hands-on experience developing 4+ full-stack and responsive web applications utilizing React.js, Node.js, Express.js, MongoDB, JavaScript, and REST APIs. Solved 300+ Data Structures and Algorithms problems on LeetCode and CodeChef across 10+ algorithmic topics with an average runtime optimization of 35%. Earned participation certificate at CP Rumble intra-college coding contest among 50+ participants. Proficient in OOP, DBMS, OS, and System Design basics, seeking a Software Engineering Internship.

EDUCATION

Indian Institute of Information Technology (IIIT) Ranchi

Nov 2024 – May 2028

B.Tech – Computer Science & Engineering (Data Science & AI) CGPA: 7.31/10

Ranchi, Jharkhand

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript (ES6+), SQL
- **Frontend Development:** React.js, HTML5, CSS3, Bootstrap, Leaflet.js, DOM Manipulation, Responsive Web Design, CSS Flexbox, CSS Grid
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, JSON Parsing, API Integration, Postman, Async/Await
- **Databases:** MongoDB, Mongoose ODM, MySQL, Firebase, Indexing
- **Tools & Cloud Platforms:** Git, GitHub, VS Code, Render, Netlify, GitHub Pages, NPM
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

PROJECTS

RescueNet – Emergency SOS & NGO Clustering Platform — *Node.js, Express.js, MongoDB, Leaflet.js, JavaScript* [GitHub](#)

- Engineered a full-stack real-time emergency dispatch system processing 100+ simulated concurrent SOS requests across 3 primary relief supply categories.
- Implemented spatial demand clustering algorithms using Leaflet.js and Haversine distance, aggregating distress signals within a 1km radius and reducing emergency routing delays by 45%.
- Configured MongoDB Atlas cloud database with **2dsphere** geospatial indexing, achieving sub-50ms query response times for location payload queries.
- Built RESTful API endpoints handling JSON payloads with full input validation and error handling across 4+ edge cases.

Weather Forecast Web Application — *React.js, REST API, JavaScript, CSS3*

[GitHub](#)

- Integrated OpenWeatherMap REST API to retrieve real-time weather metrics for 195+ countries, maintaining sub-300ms API response latency and 99.9% client-side availability.
- Engineered 6+ modular React components leveraging **useState** and **useEffect** hooks, reducing code redundancy by 40% across 3 screen breakpoints.
- Built custom error-handling workflows for 4+ HTTP failure states (404 invalid city, 408 timeout, 500 server errors), improving UI fault tolerance by 35%.

Personal Portfolio Website — *HTML5, CSS3, JavaScript, GitHub Pages*

[GitHub](#)

- Architected and deployed a 6-section responsive portfolio showcasing 5+ projects and 15+ technical skills, achieving 100/100 Google Lighthouse performance score.
- Integrated 4+ custom CSS keyframe animations and asynchronous JavaScript DOM event listeners, increasing page load speed by 25%.

Stranger Things Themed Interactive Website — *HTML5, CSS3, JavaScript*

[GitHub](#)

- Developed a 5-page interactive web application featuring 8+ CSS animations and JavaScript DOM manipulation routines.
- Tested and validated cross-browser functionality across 4 major browsers (Chrome, Firefox, Edge, Safari) and 3 device screen dimensions with 100% layout consistency.

EXPERIENCE

Competitive Programmer – LeetCode & CodeChef

Nov 2024 – Present

Self-Initiated

IIIT Ranchi

- Solved 300+ Data Structures and Algorithms problems across 10+ core topics including Graphs, Dynamic Programming, Trees, Heap, and Sorting algorithms in C++.
- Maintained an active problem-solving rate of 25+ problems/month over 6+ consecutive months, optimizing execution time complexity by 20–40% using Big-O analysis.
- Competed in **CP Rumble**, IIIT Ranchi's intra-college contest, outranking peers among 50+ participants to earn an official certificate.

ACHIEVEMENTS

- Solved **300+ DSA problems** across LeetCode and CodeChef in Year 1, mastering 10+ algorithmic paradigms including Dynamic Programming and Graph traversals.
- Built and deployed **4+ full-stack and responsive web applications** utilizing React.js, Node.js, Express.js, and MongoDB with 25+ integrated features.
- Earned Certificate of Participation at **CP Rumble** (IIIT Ranchi), placing among the top candidates in a pool of 50+ participants.
- Maintained **CGPA 7.31/10** while executing 100+ hours of full-stack project development and solving 300+ competitive programming challenges.